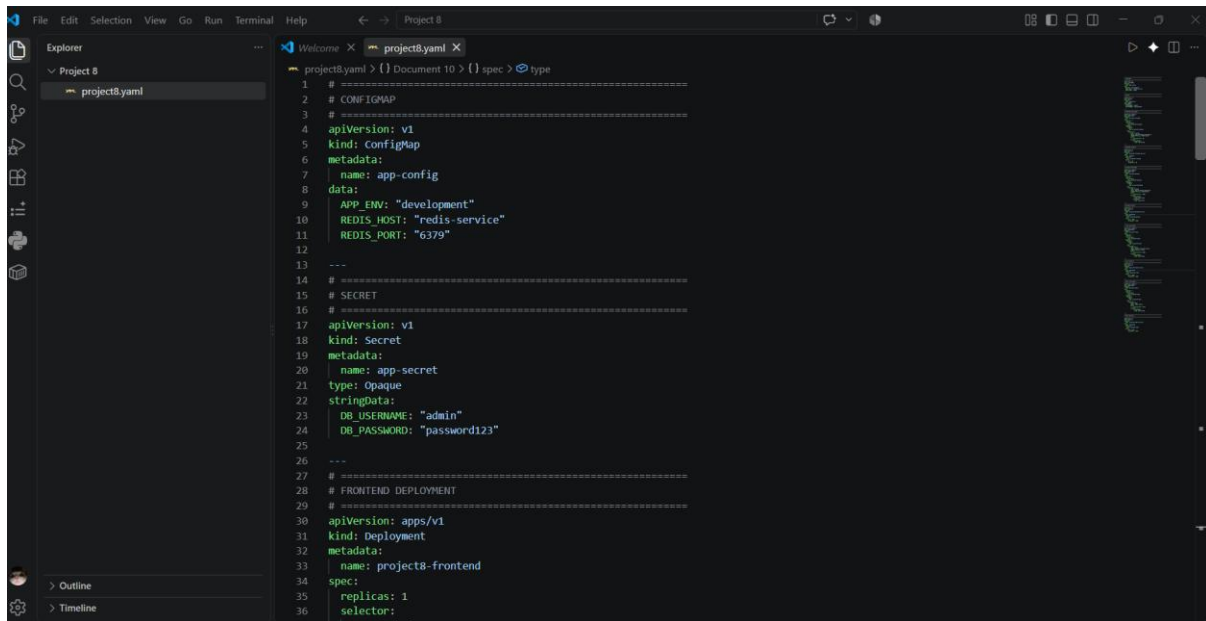


Name: Om Dhamame

PRN: 23070122155



```
1 # =====
2 # CONFIGMAP
3 # =====
4 apiVersion: v1
5 kind: ConfigMap
6 metadata:
7   name: app-config
8 data:
9   APP_ENV: "development"
10   REDIS_HOST: "redis-service"
11   REDIS_PORT: "6379"
12 ---
13 # =====
14 # SECRET
15 # =====
16 apiVersion: v1
17 kind: Secret
18 metadata:
19   name: app-secret
20 type: Opaque
21 stringData:
22   DB_USERNAME: "admin"
23   DB_PASSWORD: "password123"
24 ---
25 # =====
26 # FRONTEND DEPLOYMENT
27 # =====
28 apiVersion: apps/v1
29 kind: Deployment
30 metadata:
31   name: project8-frontend
32 spec:
33   replicas: 1
34   selector:
```

```
● PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl apply -f project8.yaml
configmap/app-config created
secret/app-secret created
deployment.apps/project8-frontend created
service/project8-frontend-service created
deployment.apps/project8-backend created
service/project8-backend-service created
deployment.apps/project8-flask created
service/project8-flask-service created
deployment.apps/project8-redis created
service/project8-redis-service created
○ PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> |
```

```

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
backend              1/1     1             1            24d
flask-deployment    2/2     2             2            35d
frontend            1/1     1             1            24d
mongo               1/1     1             1            14m
mongo-express       1/1     1             1            14m
project8-backend    1/1     1             1            39s
project8-flask      2/2     2             2            39s
project8-frontend   1/1     1             1            40s
project8-redis      1/1     1             1            39s
redis               1/1     1             1            24d
social-media-app    2/2     2             2            43m

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
backend-service     ClusterIP   10.99.250.180 <none>         5000/TCP          24d
flask-service       NodePort    10.109.217.59 <none>         5000:30007/TCP    35d
frontend-service    NodePort    10.103.10.173 <none>         80:31777/TCP      24d
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP           35d
mongo-express-service NodePort    10.108.217.44 <none>         8081:31340/TCP    14m
mongo-service       ClusterIP   10.110.253.176 <none>         27017/TCP         14m
project8-backend-service ClusterIP   10.104.155.121 <none>         5000/TCP          47s
project8-flask-service ClusterIP   10.101.61.146 <none>         5000/TCP          47s
project8-frontend-service NodePort    10.98.162.38 <none>         80:31828/TCP      47s
project8-redis-service ClusterIP   10.99.86.106 <none>         6379/TCP          47s
redis              ClusterIP   10.99.67.140 <none>         6379/TCP          24d
social-media-service NodePort    10.101.35.224 <none>         80:30405/TCP      43m

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get configmap app-config
NAME    DATA   AGE
app-config 3       51s

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get secret app-secret
NAME    TYPE    DATA   AGE
app-secret Opaque  2       53s

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-54c679dfd9-g6hq9            1/1     Running   1 (47m ago) 24d
flask-deployment-84964d4d6b-2p662   1/1     Running   2 (47m ago) 35d
flask-deployment-84964d4d6b-lbncm    1/1     Running   2 (47m ago) 35d

```

```

app-secret Opaque 2       53s

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-54c679dfd9-g6hq9            1/1     Running   1 (47m ago) 24d
flask-deployment-84964d4d6b-2p662   1/1     Running   2 (47m ago) 35d
flask-deployment-84964d4d6b-lbncm    1/1     Running   2 (47m ago) 35d
frontend-cbb6d74c4-cz2bx            1/1     Running   1 (47m ago) 24d
load-generator                       1/1     Running   0           18m
mongo-58579ff459-mtp4f              1/1     Running   0           14m
mongo-express-7cbbbdd478-46fv8      1/1     Running   0           8m33s
project8-backend-699654fd74-k8qqk    1/1     Running   0           55s
project8-flask-854f4d6789-c8pgc      1/1     Running   0           55s
project8-flask-854f4d6789-ffxmb      1/1     Running   0           55s
project8-frontend-58bc867c69-vhdb7   1/1     Running   0           55s
project8-redis-59645bf9d8-ggr6n      1/1     Running   0           55s
redis-696c967b87-d7qx2              1/1     Running   1 (47m ago) 24d
social-media-app-5475bb6f57-txnkc    1/1     Running   0           17m
social-media-app-5475bb6f57-znnd8    1/1     Running   0           43m

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8>

```

```

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl exec deployment/project8-frontend -- wget -qO- http://project8-backend-service:5000
{"message":"Hello from Flask Backend!","redis":"Connected"}
PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl exec deployment/project8-frontend -- wget -qO- http://project8-backend-service:5000
{"message":"Hello from Flask Backend!","redis":"Connected"}
PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8>

```

```

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl get endpoints project8-backend-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 Endpointslice
NAME                               ENDPOINTS                               AGE
project8-backend-service  10.244.0.29:5000                        2m40s
PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> kubectl logs deployment/project8-backend
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.244.0.29:5000
Press CTRL+C to quit

```

```

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 8> minikube service project8-frontend-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	project8-frontend-service	80	http://192.168.49.2:31828

```

🔗 Starting tunnel for service project8-frontend-service.

```

NAMESPACE	NAME	TARGET PORT	URL
default	project8-frontend-service		http://127.0.0.1:59753

```

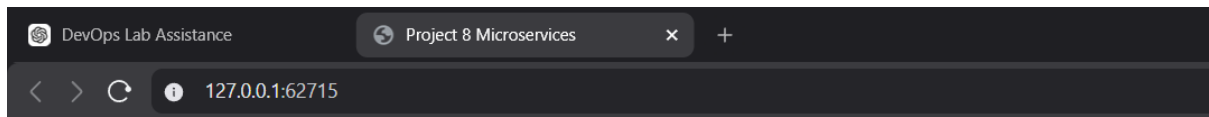
🌐 Opening service default/project8-frontend-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.

```

< > ↻ ⓘ 127.0.0.1:59753

Hello from Frontend

Loading...



Hello from Project 8 Frontend

Kubernetes Microservices Application

Check Backend

Backend status will appear here.



Hello from Project 8 Frontend

Kubernetes Microservices Application

Check Backend

Backend Connected Successfully!

nginx/1.31.4